Project Design Phase-I
Solution Architecture

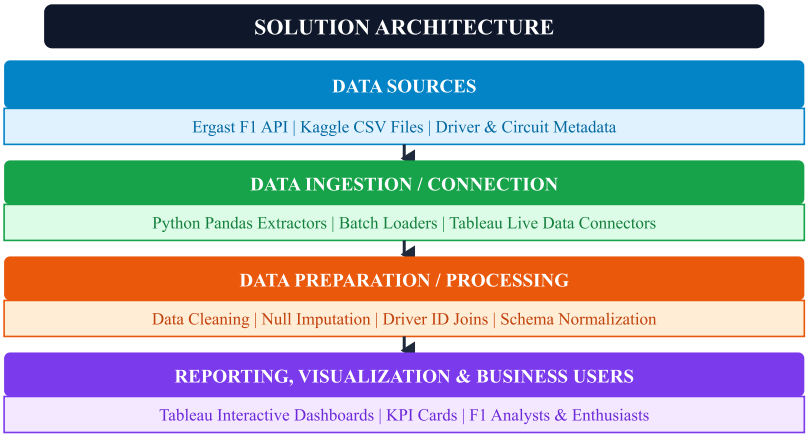
Date	31 January 2026
Team ID	anshrathore0202-cyber
Project Name	F1 Racing Intelligence Analysis
Maximum Marks	4 Marks

Solution Architecture:

Solution architecture bridges the gap between business problems and technology solutions. Its primary goals are to:

- Find the best technology solution to solve existing F1 data fragmented analysis challenges.
- Describe the structure, characteristics, behavior, and pipeline stages for all project stakeholders.
- Define ETL features, analytical layers, visualization dashboard components, and system requirements.
- Provide technical specifications according to which the solution is executed, managed, and delivered.

Solution Architecture Diagram



Component Description Table

Component Name	Description / Role in Architecture	Technology Used
Data Sources	Extract raw multi-season Formula 1 race results, lap times, pit stop durations, qualifying records, and constructor points data.	Ergast F1 API, Kaggle Datasets (CSV)
Data Ingestion Layer	Load raw telemetry files into the local workspace and establish automated data connection pipelines.	Python Pandas Ingestion Scripts, File I/O
Data Preparation & Processing	Perform batch cleaning, null imputation, timestamp formatting, joining circuit details, and creating normalized relational tables.	Python (Pandas, NumPy), VS Code Workspace
Data Storage & Data Layer	Maintain structured CSV datastores and database extracts for fast offline query execution and persistent analytical storage.	Structured CSV Datastore, SQLite DB
Analytics & BI Layer	Compute analytical measures including qualifying-to-finish position delta, stint lap pace, pit efficiency, and team dominance scores.	Tableau Desktop Calculated Fields, SQL

Component Name	Description / Role in Architecture	Technology Used
Reporting & Visualization	Deliver interactive visual dashboards, stint degradation plots, track heatmaps, and dynamic season filters for end users.	Tableau Desktop / Tableau Public
Business Users / Consumption	Motorsports analysts, sports journalists, and F1 enthusiasts who interact with dashboards to explore race performance insights.	Web Browser / Tableau Public Interface